Unit 8.01: Invariants, Monovariants, Coloring & Combinatorial Games

Q1. The coordinate diagram shows the parabolic curve f_1x = x^2 - 20x + 118 in the domain of Invariants, Monovariants, Coloring & Games. By completing the square f_1x = x - h^2 + k , find the sum k + f_112 .

[ ] A: 50
[ ] B: 22
[ ] C: 38
[x] D: 40

Solution: **Mark Scheme & Examiner Solution Invariants, Monovariants, Coloring & Games:** 1. Complete the square: x^2 - 20x + 118 = x - 10^2 - 100 + 118 = x - 10^2 + 18 , giving minimum value k = 18 . **[M1]** 2. Evaluate at x = 12 : f_112 = 2^2 + 18 = 22 . **[M1]** 3. Sum the two quantities: k + f_112 = 18 + 22 = 40 . **[A1]** - **Final Verified Answer:** 40 .

Q2. A mathematical model in Invariants, Monovariants, Coloring & Games is defined by the cubic function y_2x = 3x^3 - 10x^2 + 15x - 15 . Calculate the exact gradient of the tangent to the curve at the point where x = 3 .

[x] A: 36
[ ] B: 96
[ ] C: 21
[ ] D: 22

Solution: **Mark Scheme & Examiner Solution Invariants, Monovariants, Coloring & Games:** 1. Differentiate term by term using the power rule: frac{mathrm{d}y}{mathrm{d}x} = 9x^2 - 20x + 15 . **[M1]** 2. Substitute x = 3 : left.frac{mathrm{d}y}{mathrm{d}x}right|_{x=3} = 99 - 203 + 15 = 36 . **[M1][A1]** - **Final Verified Answer:** 36 .

Q3. In Invariants, Monovariants, Coloring & Games, two displacement vectors are given by mathbf{u}_{3} = langle 10, -8 rangle and mathbf{v}_{3} = langle 9, 7 rangle . Compute the exact scalar value of mathbf{u}_{3} cdot mathbf{v}_{3} + |mathbf{u}_{3}|^2 .

[ ] A: 34
[x] B: 198
[ ] C: 164
[ ] D: 310

Solution: **Mark Scheme & Examiner Solution Invariants, Monovariants, Coloring & Games:** 1. Compute the inner product: mathbf{u}_{3} cdot mathbf{v}_{3} = 109 + -87 = 90 + -56 = 34 . **[M1]** 2. Compute the squared norm: |mathbf{u}_{3}|^2 = 10^2 + -8^2 = 164 . **[M1]** 3. Combine: 34 + 164 = 198 . **[A1]** - **Final Verified Answer:** 198 .

Q4. The linear growth profile shown in the diagram for Invariants, Monovariants, Coloring & Games follows an arithmetic progression with first term a_1 = 11 and common difference d = 6 . Find the exact sum S_{15} of the first 15 terms.

[ ] A: 95
[ ] B: 885
[x] C: 795
[ ] D: 1425

Solution: **Mark Scheme & Examiner Solution Invariants, Monovariants, Coloring & Games:** 1. Use the sum formula for an arithmetic series: S_n = frac{n}{2}[2a_1 + n - 1d] . **[M1]** 2. Substitute n = 15, a_1 = 11, d = 6 : S_{15} = frac{15}{2}[22 + 14 times 6] = frac{15}{2} times 106 = 795 . **[M1][A1]** - **Final Verified Answer:** 795 .

Q5. In Invariants, Monovariants, Coloring & Games, evaluate the exact definite integral I_5 = int_0^{2} 18x^2 + 40x,mathrm{d}x .

[ ] A: 152
[ ] B: 168
[ ] C: 65
[x] D: 128

Solution: **Mark Scheme & Examiner Solution Invariants, Monovariants, Coloring & Games:** 1. Integrate term by term: int 18x^2 + 40x,mathrm{d}x = left[6x^3 + 20x^2right]_0^{2} . **[M1]** 2. Substitute the upper and lower limits: 62^3 + 202^2 - 0 = 48 + 80 = 128 . **[M1][A1]** - **Final Verified Answer:** 128 .

Q6. In Invariants, Monovariants, Coloring & Games, consider the quadratic function f_6x = x^2 - 22x + 211 . Express f_6x in the completed-square form x - h^2 + k and compute the exact value of k + f_618 .

[x] A: 229
[ ] B: 240
[ ] C: 139
[ ] D: 222

Solution: **Mark Scheme & Examiner Solution Invariants, Monovariants, Coloring & Games:** 1. Complete the square: x^2 - 22x + 211 = x - 11^2 - 121 + 211 = x - 11^2 + 90 , giving minimum value k = 90 . **[M1]** 2. Evaluate at x = 18 : f_618 = 7^2 + 90 = 139 . **[M1]** 3. Sum the two quantities: k + f_618 = 90 + 139 = 229 . **[A1]** - **Final Verified Answer:** 229 .

Q7. The graph illustrates the cubic model y_7x = 3x^3 - 11x^2 + 33x - 22 studied in Invariants, Monovariants, Coloring & Games. Find the instantaneous gradient frac{mathrm{d}y}{mathrm{d}x} of the tangent to the curve at x = 8 .

[ ] A: 609
[x] B: 433
[ ] C: 400
[ ] D: 83

Solution: **Mark Scheme & Examiner Solution Invariants, Monovariants, Coloring & Games:** 1. Differentiate term by term using the power rule: frac{mathrm{d}y}{mathrm{d}x} = 9x^2 - 22x + 33 . **[M1]** 2. Substitute x = 8 : left.frac{mathrm{d}y}{mathrm{d}x}right|_{x=8} = 964 - 228 + 33 = 433 . **[M1][A1]** - **Final Verified Answer:** 433 .

Q8. In Invariants, Monovariants, Coloring & Games, two displacement vectors are given by mathbf{u}_{8} = langle 11, -7 rangle and mathbf{v}_{8} = langle 8, 5 rangle . Compute the exact scalar value of mathbf{u}_{8} cdot mathbf{v}_{8} + |mathbf{u}_{8}|^2 .

[ ] A: 53
[ ] B: 170
[x] C: 223
[ ] D: 293

Solution: **Mark Scheme & Examiner Solution Invariants, Monovariants, Coloring & Games:** 1. Compute the inner product: mathbf{u}_{8} cdot mathbf{v}_{8} = 118 + -75 = 88 + -35 = 53 . **[M1]** 2. Compute the squared norm: |mathbf{u}_{8}|^2 = 11^2 + -7^2 = 170 . **[M1]** 3. Combine: 53 + 170 = 223 . **[A1]** - **Final Verified Answer:** 223 .